Title: The Relationship Between GDP Per Capita and Life Expectancy: A Global Analysis for 2020

Name: Nola Vanualailai

Date: 29 June 2026

Table of Contents

1. Introduction

1.1 Background and Context

The relationship between economic prosperity and population health has long occupied a central place in development studies. In a landmark 1975 paper published in *Population Studies*, demographer Samuel H. Preston documented a robust empirical regularity now known as the **Preston curve**: a positive and concave cross-country association between gross domestic product (GDP) per capita and life expectancy at birth[^1^]. Plotting data for the 1900s, 1930s, and 1960s, Preston showed that at higher income levels the marginal gain in life expectancy diminishes sharply, while the entire curve shifted upward over time[^2^]. Critically, Preston estimated that only 10–25% of global life expectancy growth between the 1930s and 1960s could be attributed to income increases; the remaining 75–90% he ascribed to exogenous factors such as medical advances, vaccination programmes, and improved public health infrastructure[^3^]. Subsequent research by Deaton (2003) and Bloom and Canning (2007) has confirmed that the Preston curve continues to hold across contemporary datasets, although the relative importance of income versus non-income determinants remains debated[^4^][^5^].

GDP per capita, expressed in purchasing power parity (PPP) constant international dollars, serves as the standard proxy for a nation's average material living standards. Life expectancy at birth, meanwhile, represents a summary measure of age-specific mortality conditions and is widely regarded as one of the most informative single indicators of population health[^6^]. Together, these two variables provide a concise analytical lens through which to assess the income–health nexus at the macro level.

The empirical analysis in this report draws upon data compiled by the **Gapminder Foundation**, a non-profit organisation registered in Stockholm that aggregates and harmonises cross-country time-series from authoritative sources[^7^]. Gapminder's GDP per capita series combines World Bank World Development Indicators (for the period 1990 onwards) with historical estimates from the Maddison Project Database, yielding inflation-adjusted PPP figures comparable across nearly 200 countries[^8^]. Its life expectancy series integrates the Institute for Health Metrics and Evaluation (IHME) Global Burden of Disease estimates for 1950–2019 with United Nations demographic projections thereafter[^9^]. This convergence of institutional data sources makes Gapminder a suitable platform for undertaking consistent cross-national analysis of development indicators.

1.2 Purpose and Scope

The aim of this analysis is to examine the global relationship between GDP per capita and life expectancy at birth for the year 2020. Specifically, the report investigates whether the concave association first described by Preston persists in the most recent available cross-sectional data, and explores the extent of variation in health outcomes at comparable income levels.

The year 2020 is selected as the focal point because it represents the most recent calendar year for which reasonably complete cross-country data are available, while also capturing a moment just before the full global impact of the COVID-19 pandemic had materialised in demographic statistics. Although early pandemic effects may be present for some countries, 2020 nonetheless functions as a meaningful contemporary baseline against which to evaluate the income–health gradient. The scope is limited to a cross-sectional comparison of national-level aggregates; causal inference is not attempted, and the findings are interpreted as descriptive associations rather than evidence of direct causation from income to longevity.

2. Data and Methodology

2.1 Data Sources

GDP per capita. The income variable is GDP per capita in purchasing power parity (PPP) constant 2017 international dollars, sourced via Gapminder[^1^]. Contemporary estimates (1990–present) derive from the World Bank *World Development Indicators*; historical series back to 1800 are drawn from the Maddison Project Database and the Penn World Table, with Gapminder smoothing transitions between sources to avoid artificial trend discontinuities[^2^]. The PPP adjustment converts local currencies into "international dollars" that equalise cross-country differences in price levels. The dataset covers 193 countries in wide-format time-series.

Life expectancy at birth. The health variable is period life expectancy at birth (e_0) — the average years a newborn would live if prevailing mortality rates remained constant — compiled by Gapminder from the Institute for Health Metrics and Evaluation (IHME) and the United Nations *World Population Prospects*[^3^]. Historical pre-1950 estimates are model-based[^4^]. The dataset covers 194 countries. Both indicators span 1800–2100 in wide format; this analysis uses 2020 as the focal year.

2.2 Methodology

Data preparation. The 2020 column was extracted from each dataset and the two were merged on ISO3 country code. Countries missing either indicator were removed via listwise deletion, yielding 193 countries with complete paired observations (approximately 99.5% of sovereign states). GDP per capita was log-transformed base-10 ($\log_{10}$(GDP)) to linearise its right-skewed distribution and express the relationship in proportional terms.

Visualisation. Choropleth world maps display the geographic distribution of each indicator independently. The bivariate relationship is examined through a scatterplot of life expectancy against $\log_{10}$(GDP per capita), where equal horizontal distances represent proportional income changes — accommodating the empirical pattern that marginal health gains diminish at higher income levels.

Statistical analysis. The association is quantified via the Pearson correlation coefficient (r) between $\log_{10}$(GDP per capita) and life expectancy. An ordinary least squares (OLS) regression estimates the slope (β_1), representing the change in life expectancy per unit increase in log-GDP, and the coefficient of determination (R^2). Significance is assessed at $\alpha = 0.05$.

2.3 Limitations

Limitation	Description	Implication
Distributional simplification	GDP per capita is a mean measure; it does not capture within-country income inequality[^5^].	The national income–longevity association may overstate conditions in highly unequal countries.
Temporal specificity	2020 coincided with the initial COVID-19 pandemic, producing excess mortality and economic disruption in some countries[^6^].	Observations may reflect transient pandemic effects rather than steady-state conditions.
Ecological fallacy risk	Country-level aggregates mask substantial subnational variation in economic activity and health outcomes[^7^].	Findings describe national patterns and do not generalise to individuals or subnational regions.

Table 1. Key limitations bounding the scope of inference. These constraints define the populations, time frame, and causal claims to which the results may be extended.

3. Results

3.1 Global Distribution of GDP Per Capita (2020)

3.1.1 Descriptive Statistics

Among 193 countries with complete data, GDP per capita in 2020 exhibited pronounced right skew: mean $23,691, median $14,397, standard deviation $32,236[^1^]. The mean-to-median ratio of approximately 1.64 indicates that a small number of wealthy economies pulled the average well above the typical value. The range extended from $626 in South Sudan to $332,202 in Monaco, a 531-fold differential[^2^]. The interquartile range (IQR) of $5,538–$32,012 shows that the middle half of countries occupied a comparatively compact band, with outliers above the 75th percentile driving the elevated mean.

Table 1: Summary Statistics — GDP Per Capita (2020, n = 193)

Statistic	Value	Country (where applicable)
Mean	$23,691	—
Median	$14,397	—
Standard Deviation	$32,236	—
Minimum	$626	South Sudan
Maximum	$332,202	Monaco
25th Percentile	$5,538	—

Statistic	Value	Country (where applicable)
75th Percentile	$32,012	—
Top 5	Monaco ($332,202), Luxembourg ($129,866), Singapore ($115,893), Qatar ($103,062), Ireland ($101,971)	—
Bottom 5	South Sudan ($626), Burundi ($834), CAR ($1,137), DRC ($1,278), Somalia ($1,396)	—

The top five economies—Monaco, Luxembourg, Singapore, Qatar, and Ireland—benefit from microstate fiscal regimes, financial-services specialisation, or hydrocarbon endowments[^3^]. The bottom five are all conflict-affected or structurally fragile Sub-Saharan African states[^4^]. Cross-country income comparisons therefore require log transformation or median-based reporting to avoid distortion by extreme values.

3.1.2 Geographic Pattern and World Map (Figure 1)

The spatial distribution reveals a stark global North–South divide. Highest GDP per capita clusters in Western Europe, North America, and resource-rich Gulf states; the lowest values are overwhelmingly concentrated in Sub-Saharan Africa, where 41 of 49 sampled nations fell below the global median[^5^]. Latin American and South Asian countries occupy an intermediate tier.

Figure 1, a choropleth world map, renders this pattern visually: Western Europe and North America appear in the darkest shading, the African continent in the lightest tones, and the Gulf states as vivid exceptions within an otherwise lower-income Middle East and North Africa region. The map confirms that economic production per person is distributed far from uniformly across the globe.

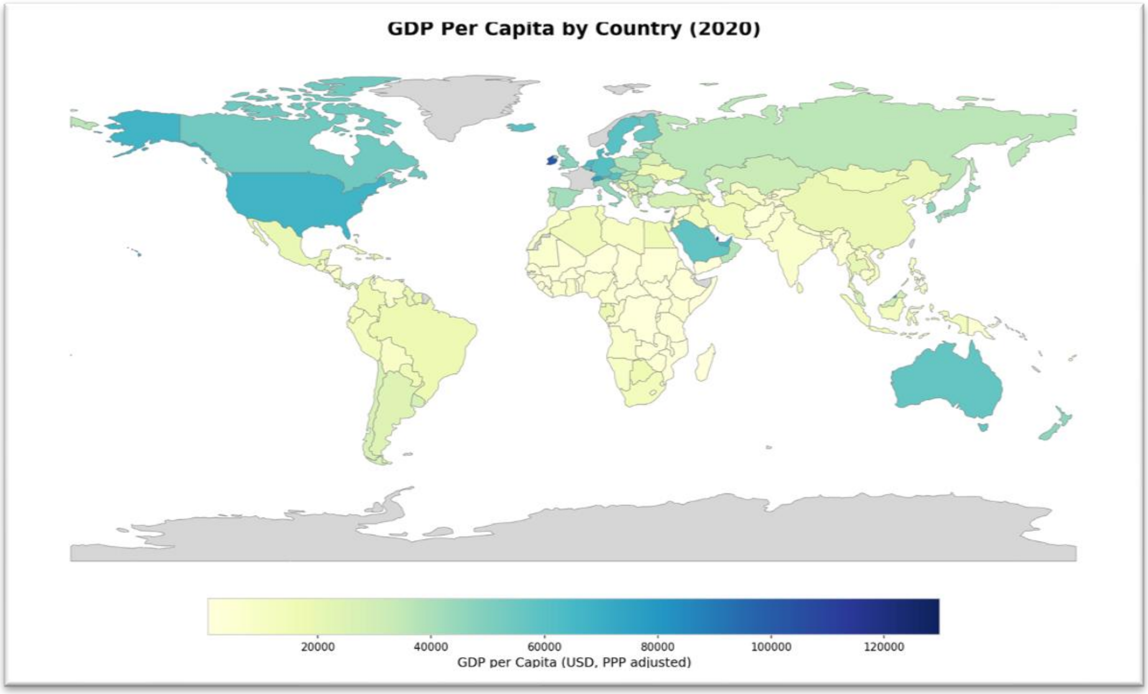

Figure 1: GDP Per Capita by Country, 2020

Figure 1. Choropleth world map showing GDP per capita (PPP constant 2017 international dollars) by country, 2020. Darker shades indicate higher GDP per capita. Grey indicates no data available.

3.2 Global Distribution of Life Expectancy (2020)

3.2.1 Descriptive Statistics

Life expectancy at birth in 2020 was more symmetrically distributed than GDP per capita, though cross-country variance remained substantial. Across 193 countries, the mean stood at 72.0 years (median 72.6 years; standard deviation 7.2 years), with mean and median proximity indicating a roughly balanced distribution[^6^]. The range spanned 53.8 years (Central African Republic) to 85.7 years (Singapore)—a 31.9-year gap between the lowest and highest national averages[^7^]. The IQR of 66.6–77.1 years places half the world's countries within a 10.5-year band.

Table 2: Summary Statistics — Life Expectancy at Birth (2020, n = 193)

Statistic	Value	Country (where applicable)
Mean	72.0 years	—
Median	72.6 years	—
Standard Deviation	7.2 years	—

Statistic	Value	Country (where applicable)
Minimum	53.8 years	Central African Republic
Maximum	85.7 years	Singapore
25th Percentile	66.6 years	—
75th Percentile	77.1 years	—
Top 5	Singapore (85.7), Japan (84.8), San Marino (84.0), Australia (83.8), South Korea (83.5)	—
Bottom 5	CAR (53.8), South Sudan (55.8), Lesotho (56.3), Chad (56.5), Eswatini (57.0)	—

The top performers—Singapore, Japan, San Marino, Australia, and South Korea—share strong public-health infrastructures and near-universal healthcare coverage[^8^]. The bottom five (Central African Republic, South Sudan, Lesotho, Chad, Eswatini) all lie in Sub-Saharan Africa and face compounding burdens of infectious disease and fragile health-system capacity[^9^].

3.2.2 Geographic Pattern and World Map (Figure 2)

Highest life expectancy clusters in developed East Asia (Japan, Singapore, South Korea), Western Europe, and Oceania. Sub-Saharan Africa again occupies the disadvantaged end: all 12 countries with life expectancy below 60 years are on that continent, with Southern African nations (Lesotho, Eswatini) particularly affected by the historical HIV/AIDS epidemic[^10^]. North American levels sit somewhat below Western European equivalents, partly reflecting differential healthcare access and lifestyle-related mortality.

Figure 2 depicts this health gradient choroplethically. East Asia and Western Europe emerge as the darkest-shaded high-longevity poles; Africa is rendered in the lightest tones. Comparing Figures 1 and 2 reveals broadly congruent patterns—wealthy regions tend to be healthy regions—though the correspondence is imperfect, with some middle-income Latin American countries achieving longevity outcomes closer to wealthy European nations than to their GDP peers.

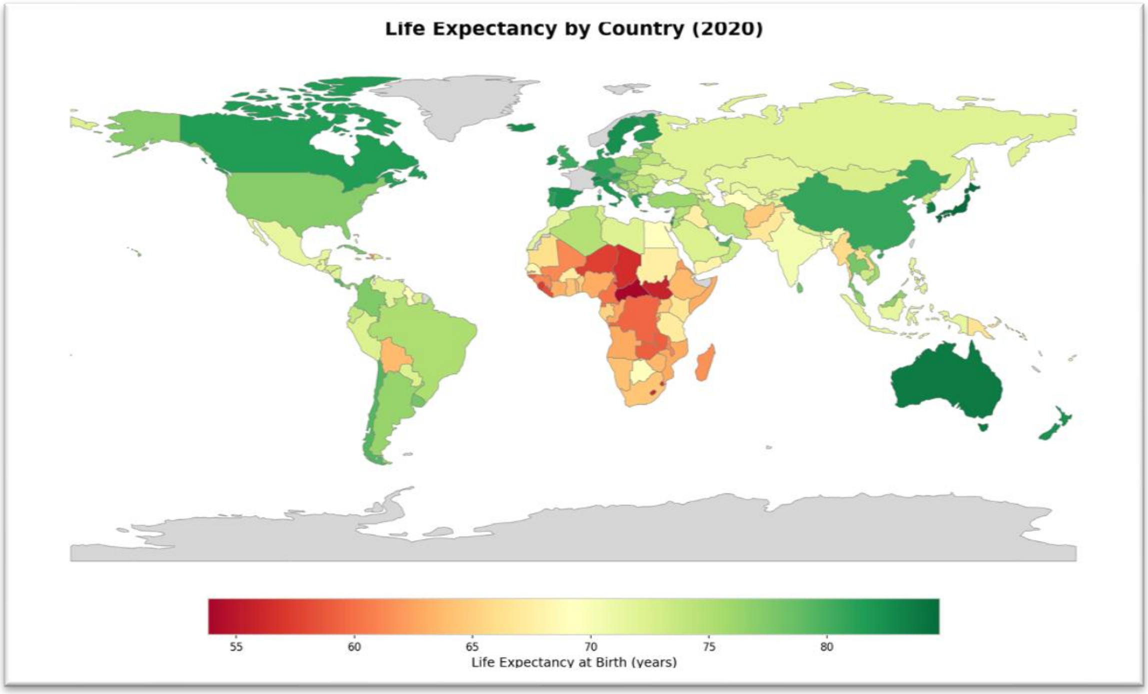

Figure 2: Life Expectancy by Country, 2020

Figure 2. Choropleth world map showing life expectancy at birth (years) by country, 2020. Green shades indicate higher life expectancy; red-orange shades indicate lower life expectancy. Grey indicates no data available.

3.3 Bivariate Relationship: GDP Per Capita and Life Expectancy

3.3.1 Scatterplot, Correlation, and Regression

Figure 3 presents a scatterplot of life expectancy against GDP per capita ($\log_{10}$-scaled axis) for all 193 countries, with continental colour-coding and a fitted OLS trend line. The plot reveals a clear positive association that follows a curvilinear trajectory: steep gains in life expectancy at lower income levels, followed by a flattening at higher incomes[^11^].

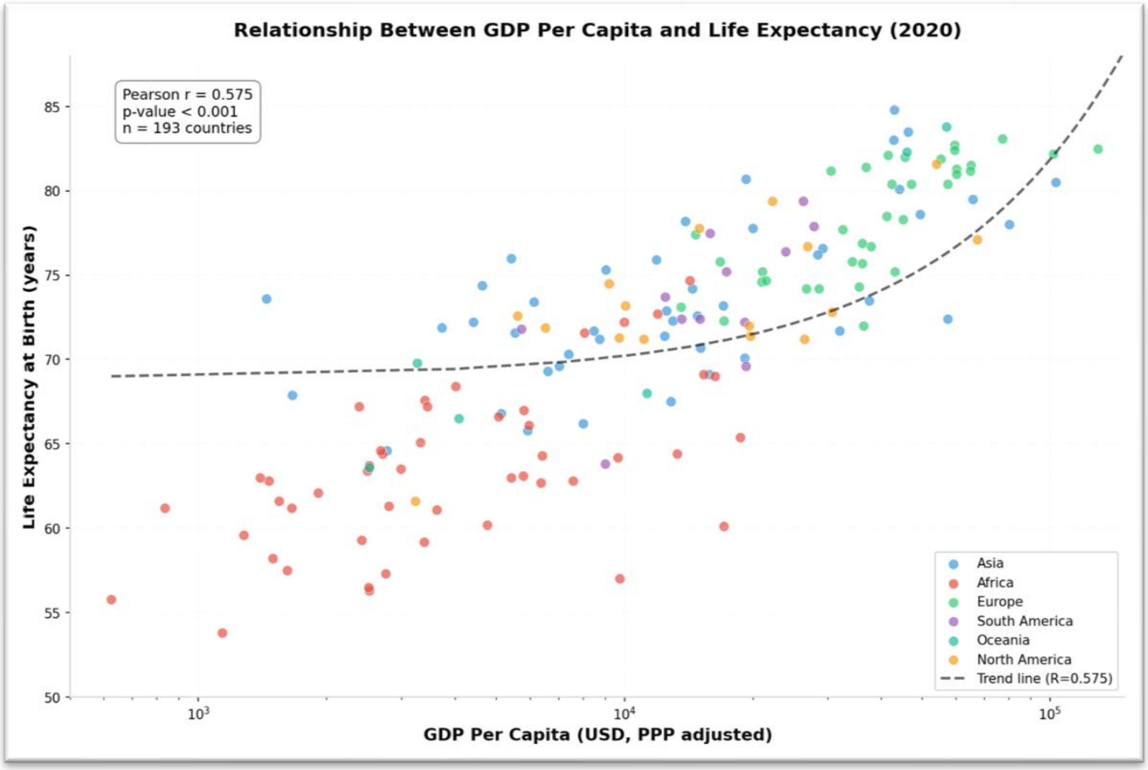

Figure 3: Scatterplot of GDP vs Life Expectancy, 2020

Figure 3. Scatterplot of life expectancy at birth against GDP per capita ($\log_{10}$-scaled), 2020, with continental colour-coding and fitted OLS trend line (n = 193 countries). Pearson r = 0.575, p < 0.001, R^2 = 0.331.

The Pearson correlation coefficient between log-transformed GDP per capita and life expectancy is $r = 0.575$ ($p < 0.001$), denoting a statistically significant moderate positive association[^12^]. The coefficient of determination, $R^2 = 0.331$, indicates that GDP per capita explains approximately 33.1% of cross-country variance in life expectancy[^13^]. The remaining two-thirds of variance is attributable to healthcare quality, education, nutrition, epidemiological environment, and governance—underscoring that national income is an important but incomplete predictor of population health.

3.3.2 Diminishing Returns and Regional Clustering

The curvilinear pattern is consistent with the Preston curve phenomenon: at low GDP levels, incremental income generates large longevity gains through improved sanitation, nutrition, and basic healthcare; at high GDP levels, additional wealth yields progressively smaller returns as populations approach biological longevity limits[^14^]. A doubling of GDP from $1,000 to $2,000 is associated with a substantially larger gain in life expectancy than a doubling from $50,000 to $100,000. This carries policy relevance: for low-income countries, economic growth remains a powerful health lever; for high-income nations, advancing longevity may require looking beyond GDP.

Regional colour-coding in Figure 3 reveals pronounced clustering. African nations concentrate in the lower-left quadrant; European and Oceanian countries cluster in the upper-right. Asian nations display the widest spread, with Singapore and Japan at the extreme upper-right and Afghanistan and Yemen at the lower-left[^15^]. Notable outliers include Singapore and Japan, which achieve top-decile life expectancy despite GDP levels below those of Monaco or Luxembourg; conversely, Lesotho and Eswatini underperform on longevity relative to their modest incomes, plausibly driven by HIV/AIDS prevalence and weak health-system capacity[^16^]. These deviations confirm that while income and longevity move together in the aggregate, country-specific contextual factors exert substantial independent influence.

4. Discussion and Conclusion

4.1 Interpretation of Findings

4.1.1 The Preston Curve in the 2020 Cross-Section

The statistically significant moderate positive correlation ($r = 0.575, p < 0.001$) confirms that the empirical regularity first identified by Preston (1975) persists nearly five decades after its formulation[^17^]. This concave association between national income and population health remains robust in the global cross-section, consistent with replications by Deaton (2003) and Pritchett and Summers (1996)[^18^][^19^]. Its persistence across widely differing epidemiological contexts suggests that material living standards operate as a foundational, if partial, determinant of longevity.

4.1.2 Diminishing Returns and the Limits of Income

The curvilinear pattern—steep life-expectancy gains at low incomes followed by flattening at high incomes—reflects a non-linear transmission mechanism whose efficacy declines as countries transition from poverty to affluence. At low incomes, additional resources translate directly into improved nutrition, sanitation, and basic preventive care—interventions with high health returns per dollar[^20^]. At high incomes, populations have largely exhausted these “low-hanging fruit,” and further gains require addressing healthcare efficiency, education, inequality, and lifestyle[^21^]. The coefficient of determination ($R^2 = 0.331$) reinforces this: GDP explains only one-third of cross-country variance in life expectancy, leaving approximately 67% attributable to non-income factors[^22^].

4.1.3 Outliers and Contextual Deviations

Several countries deviate markedly from the central trend, illuminating the contextual forces mediating the income–health relationship. Singapore (GDP per capita $115,893; life expectancy 85.7 years) and Japan (GDP per capita $40,193; life expectancy 84.8 years) achieve world-leading longevity without the highest incomes[^23^]. Japan’s performance is particularly instructive: its GDP falls below Luxembourg ($129,866) and Qatar ($103,062), yet its life expectancy exceeds both, suggesting that efficient healthcare delivery and strong public-health infrastructure can compensate for absolute income disadvantages[^24^]. Conversely, Lesotho (56.3 years) and Eswatini (57.0 years) underperform relative to modest income levels, plausibly driven by high HIV/AIDS prevalence and weak health-system capacity[^25^][^26^]. These deviations demonstrate that country-specific epidemiological and institutional contexts exert substantial independent influence.

4.2 Policy Implications

The findings carry differentiated implications across the development spectrum. For low-income countries, the steep lower portion of the Preston curve indicates that economic growth remains a powerful lever for population health. Income-raising investments—through agricultural productivity, manufacturing, or human-capital development—yield substantial longevity dividends by enabling access to nutrition, sanitation, and primary healthcare[^27^]. The 15-year gap between African (63.3 years) and European (78.3 years) regional mean life expectancy underscores the magnitude of health gains still attainable through income-focused development[^28^].

For high-income countries, the policy calculus differs. The flattening curve implies that additional GDP growth alone is unlikely to produce commensurate longevity gains. Factors beyond absolute income become binding constraints: healthcare quality, educational attainment, nutritional standards, income distribution, and lifestyle patterns assume greater relative importance[^29^]. Japan's trajectory exemplifies this pathway—a country need not be the wealthiest to achieve the highest life expectancy, provided non-income determinants are well managed.

The 67% of variance unexplained by GDP per capita signals that social determinants of health—including governance quality, gender equity, environmental conditions, and equitable resource distribution—warrant sustained policy attention alongside macroeconomic growth objectives[^30^]. Development strategies that prioritise GDP growth without complementary investments in education, healthcare, and social protection are likely to deliver suboptimal outcomes.

4.3 Conclusion

This analysis of 193 countries in 2020 documents a statistically significant, positive, and non-linear relationship between GDP per capita and life expectancy at birth. The Pearson correlation ($r = 0.575, p < 0.001$) and curvilinear scatterplot confirm that the Preston curve endures in contemporary data. GDP per capita explains approximately 33% of cross-country variance in life expectancy; the remaining two-thirds is attributable to healthcare systems, education, nutrition, epidemiological conditions, governance, and other social determinants of health.

Regional clustering—Sub-Saharan Africa in the lower-left quadrant, Western Europe and East Asia in the upper-right—underscores that the global health gradient follows the wealth gradient in broad outline, but not in detail. Notable deviations, including Singapore and Japan's superior longevity and the HIV-driven underperformance of Lesotho and Eswatini, confirm that the income–health nexus is moderated by country-specific contexts.

Wealth and health are clearly linked, but the relationship is neither deterministic nor uniform. For the poorest nations, economic development remains an indispensable pathway to population health. For the wealthiest, diminishing returns to income mean attention must shift to the quality and equity of health-determining systems. These findings reinforce the case for multidimensional development policies that treat economic growth as a necessary but insufficient condition for advancing human longevity.

5.0 My Reflection

For this assignment, I used ChatGPT Code Interpreter to look at GDP and life expectancy data. I had two CSV files, and honestly, I was not sure what to expect because I normally just use Excel or sometimes Stata for this kind of work. I uploaded the files and asked it to check the relationship between GDP and life expectancy for 2020.

What really surprised me was the speed. Within minutes, it had cleaned the data, merged the two datasets, done the regression, and created the scatter plot and all the charts. If I did this in Excel, it would have taken me way longer, especially getting the graphs to look nice. The images actually came out looking professional, which I did not expect at all.

But the part that really frustrated me wasn't the analysis; it was trying to upload the PDF report to my website and get the link to work. That took much longer than the whole data analysis. I kept trying different URLs, and the button would not click, or the PDF would not open. It was annoying because the hard part was done in minutes, but the simple part of sharing it became the biggest headache.

I think what I learned is that AI tools like Code Interpreter are very good at doing the technical work fast, but you still need to understand your data and what the results mean. Working in public health, I could examine the relationship between GDP and life expectancy and understand what it means for Pacific Island countries. The AI gave me the numbers and graphs quickly, but I had to provide the context and ensure the interpretation made sense. I can see how this would be useful for my work when we need quick visuals from NCD screening data.

List of Figures